

SESSION 7

1. Restaurant constraints

NOT NULL requires values; PRIMARY KEY is unique and cannot be NULL.

```
CREATE TABLE Restaurant (  
  id INT PRIMARY KEY, name VARCHAR(100) NOT NULL,  
  location VARCHAR(100) NOT NULL, cuisine VARCHAR(60)  
);
```

2. FoodOrder foreign key

A FoodOrder must reference a valid Restaurant row.

```
CREATE TABLE FoodOrder (  
  order_id INT PRIMARY KEY, restaurant_id INT NOT NULL, user_id INT NOT NULL,  
  order_total DECIMAL(10,2) NOT NULL,  
  FOREIGN KEY (restaurant_id) REFERENCES Restaurant(id)  
);
```

3. SpotifyUser constraints

UNIQUE prevents duplicate usernames; email is required.

```
CREATE TABLE SpotifyUser (  
  user_id INT PRIMARY KEY, username VARCHAR(80) UNIQUE,  
  email VARCHAR(120) NOT NULL, subscription_type VARCHAR(30)  
);
```

4. TopSpendersView

The view joins users and orders, then exposes only orders over 1000.

```
CREATE VIEW TopSpendersView AS  
SELECT u.username, o.order_total  
FROM SpotifyUser u JOIN FoodOrder o ON u.user_id=o.user_id  
WHERE o.order_total > 1000;
```

5. Query only the view

This reads from the view rather than the underlying FoodOrder table.

```
SELECT username, order_total  
FROM TopSpendersView  
WHERE order_total > 2000  
ORDER BY order_total DESC;
```